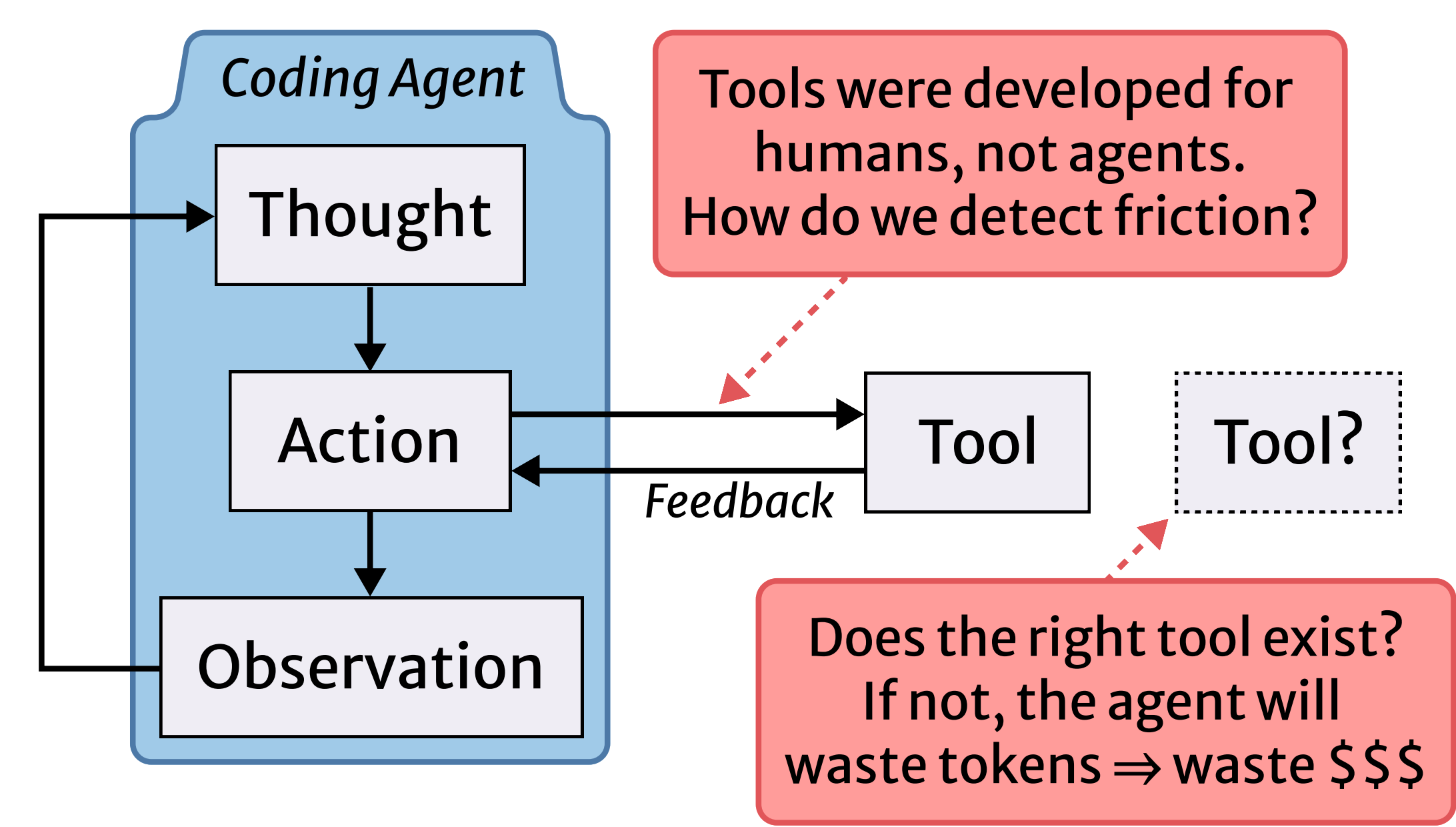


# What Do AI Coding Agents Need?

## Mining Trajectories to Identify Gaps in Language Tooling

Tommy M<sup>c</sup>Michen



T: "There's a NameError for 'BaseModel', it must not be imported. Let me add the import."  
A: [edit file — adds: from pydantic import BaseModel]  
A: python src/main.py  
O: NameError: name 'BaseModel' is not defined

T: "Still failing. Maybe it's from a different module."  
A: [edit file — changes to: from sqlalchemy import BaseModel]  
A: python src/main.py  
O: ImportError: cannot import name 'BaseModel' from 'sqlalchemy'

T: "Let me check what's actually available in the codebase."  
A: grep -rn "BaseModel".  
O: No matches found.

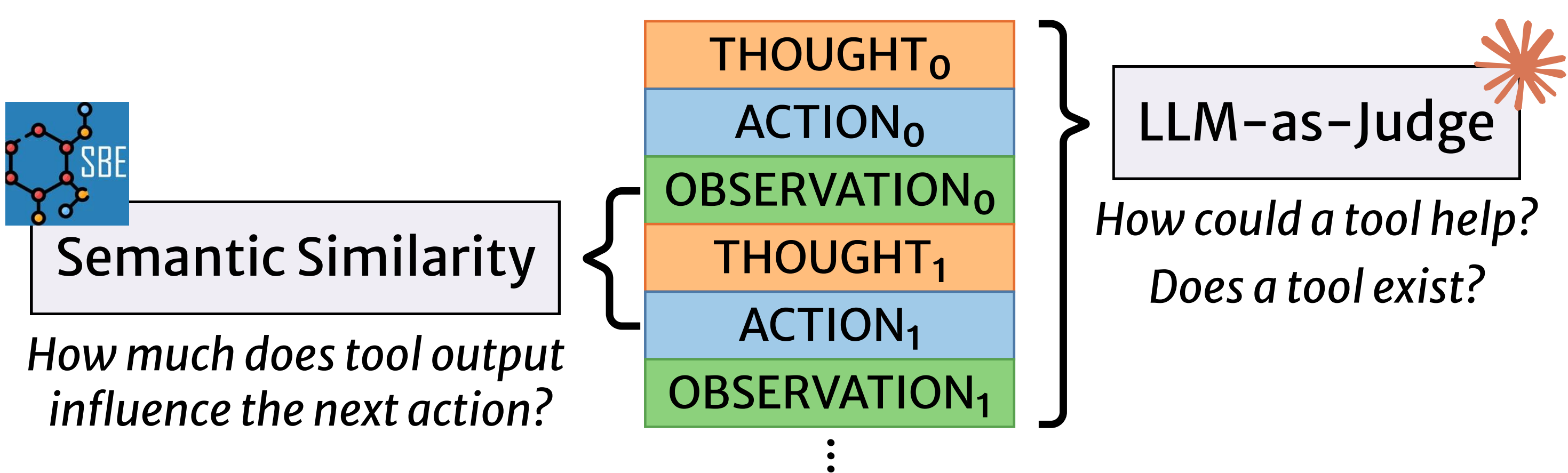
Trajectories provide rich insights

Automatic Trajectory Analysis with **tern**

How can we use them to build tools?

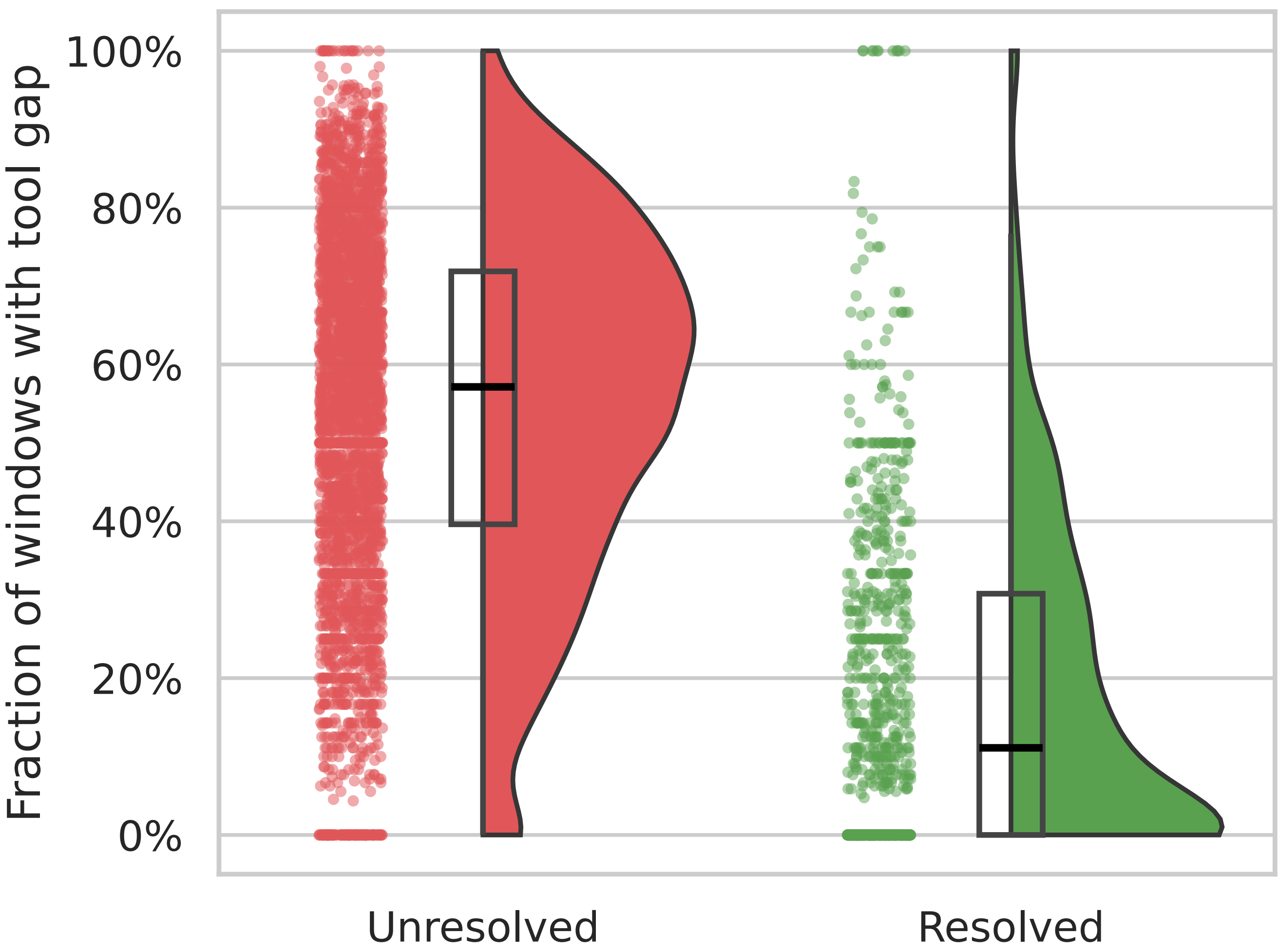


### How can we systematically identify where tools fail agents at scale?

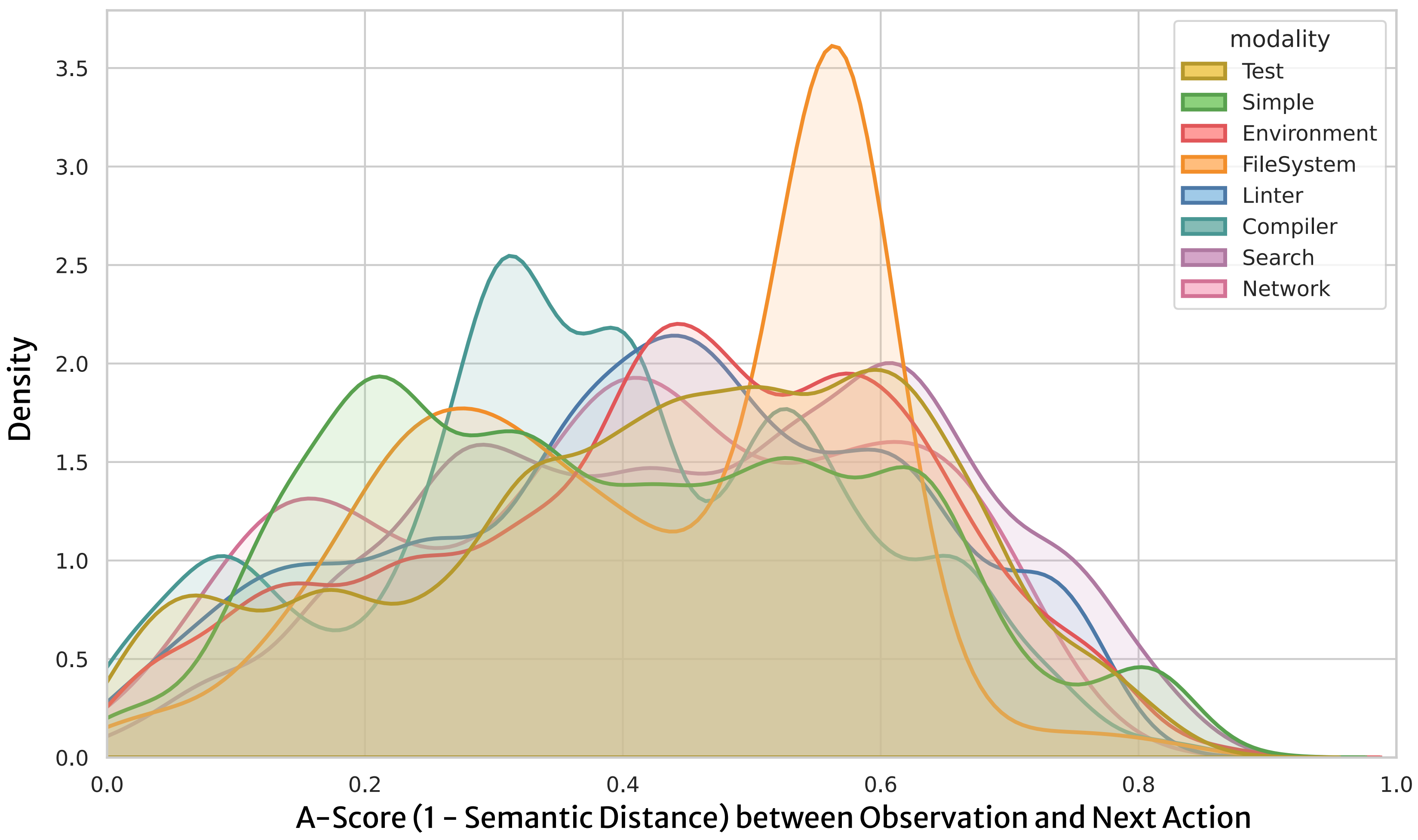


| Category         | Description                                         |
|------------------|-----------------------------------------------------|
| Syntax           | lexical errors, indentation, brackets               |
| Name Resolution  | undefined names, missing imports                    |
| Type Semantics   | type mismatches, wrong attributes                   |
| Data Flow        | variable scope, closures, use-before-def            |
| Control Flow     | missing return paths, exception handling, dead code |
| Domain Semantics | format strings, API contracts, DSL validation       |
| Runtime Behavior | requires execution to detect                        |

Tooling gaps predict failure



Tools vary in alignment with intended action



### Multiple techniques can identify a variety of tooling gaps.

| Category         | Windows | % of Total | Success Rate |
|------------------|---------|------------|--------------|
| Domain Semantics | 4,408   | 25.3%      | 4.0%         |
| Type Semantics   | 3,517   | 20.2%      | 5.5%         |
| Name Resolution  | 3,243   | 18.6%      | 3.4%         |
| Syntax           | 3,307   | 19.0%      | 2.1%         |
| Data Flow        | 1,851   | 10.6%      | 4.1%         |
| Control Flow     | 467     | 2.7%       | 4.5%         |
| Runtime Behavior | 640     | 3.7%       | 5.5%         |
| Total            | 17,433  | 100%       | 3.9%         |

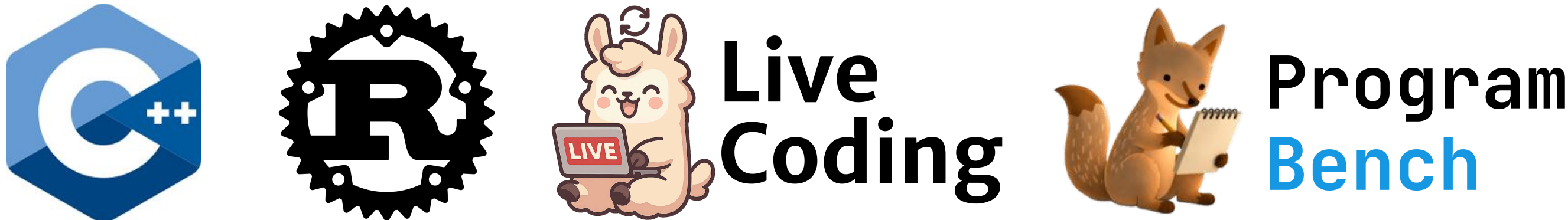
91.6% of trajectories need existing tools



Diverse opportunities for new tools

Program slicing, Call graph, Domain-specific

New benchmarks and tasks to explore



Agents exhibit a diverse set of failure modes that could be helped by new and existing tools